

“... easier to use, quicker and of less hindrance to the user... as well as cheaper...”

FROM A LETTER MENTIONING SNAPHANCES TO THE VENETIAN AMBASSADOR IN ENGLAND
WRITTEN BY THE DOGE AND SENATE OF VENICE, NOVEMBER 6, 1613

touchhole relayed the ignition flash to the breech of the barrel, firing the main gunpowder charge.

The snaphance remained popular in parts of Europe until the 19th century but, while regional styles existed, the greatest influence on its design came from France. In the late 1600s, French gunmakers published design books depicting fashionable shapes for components and their decoration. Many gunmakers in western Europe adopted these enthusiastically.

The design of the snaphance was simplified to create the first true flintlock, in which the separate pan cover and steel were combined to create a part called the frizzen. This opened when struck by the flint (see p.303). Uniting these parts into a single piece made the flintlock cheaper to manufacture and far more reliable. The flintlock had far fewer

parts than the wheel-lock—a late 17th-century flintlock might have just 16 parts compared to a wheel-lock's 40. This simplicity of design allowed flintlocks to be built more quickly.

THE FLINTLOCK IN USE

All three gunlocks—the matchlock, wheel-lock, and flintlock—remained in use throughout the 17th century, but the advantages of the flintlock were obvious. By the early 18th century, it had

▼ FLINTLOCKS IN WAR

By the 18th century, the flintlock musket was the main infantry weapon in Europe and North America, and featured prominently in the American Revolutionary War. At the Battle of Brandywine in 1777, American troops put up a stiff resistance before being defeated by British forces. Seen here are American soldiers firing their flintlock muskets in volleys.

been adopted widely. For the armies, it was cost-effective technology that could be applied toward manufacturing firearms in large numbers to standardized patterns. Gunmakers could fit a flintlock to all kinds of firearms, from a cavalry pistol to an artillery piece. Guns now became affordable for the civilian population, too. The flintlock provided travelers with useful firearms for self-defense, sportsmen with guns which were both efficient and fashionable, and duelists with weapons of deadly reliability.

Refinement of the flintlock technology continued into the 19th century, but even in its most efficient form, it had its drawbacks. Smoke produced by flintlock weapons could alert game to the presence of a hunter. The flint needed to be kept in precisely the right shape and place, and the touchhole needed to be kept clear of residue. The mechanism's exposed priming made it susceptible to bad weather. Gunmakers tried to keep the mechanism waterproof by designing a raised rib around the pan to keep out moisture, but this did not work completely. The solution to these problems came in the form of gunlocks using chemicals called fulminates (see p.80) as primers. Chemical ignition systems heralded a new era for firearms development.

AFTER >>

The flintlock mechanism continued to be used into the 1850s, but gradually gave way to a more reliable firing mechanism—the percussion cap (see pp.80–81)—which rendered it obsolete.

- **FLINTLOCK MUSKETS** were produced en masse in the late 17th century to equip armies in Europe. Large-scale military firearms production became possible in the early 18th century, and standardized patterns of flintlock weapons became available to the armies.

- **FLINTLOCK PISTOLS** were used widely as weapons for self-defense and in dueling in the 18th century. These firearms continued to be standardized into the 19th century, resulting in plain-looking mass-produced guns.

- **PERCUSSION CAPS** began replacing the flintlock in most of Europe by the 1830s. Flintlock weapons were gradually upgraded by converting them to employ percussion caps.



THE PERCUSSION CAP MECHANISM

